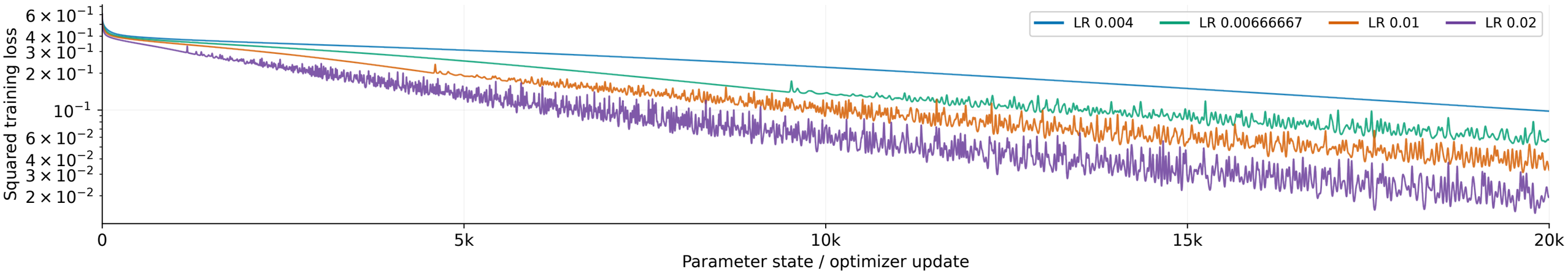
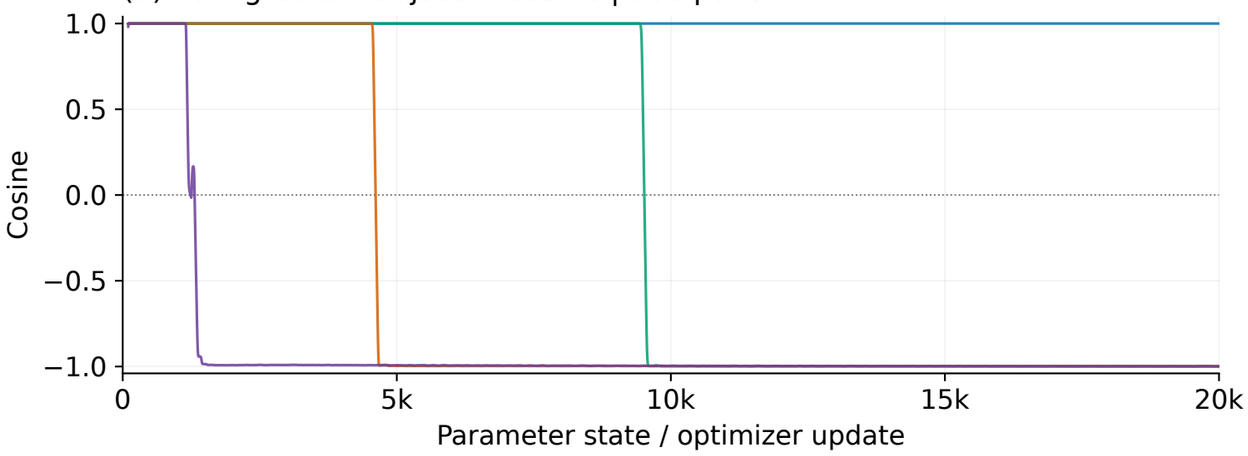
CIFAR-10 | GD geometry | four learning rates | 20,000 updates

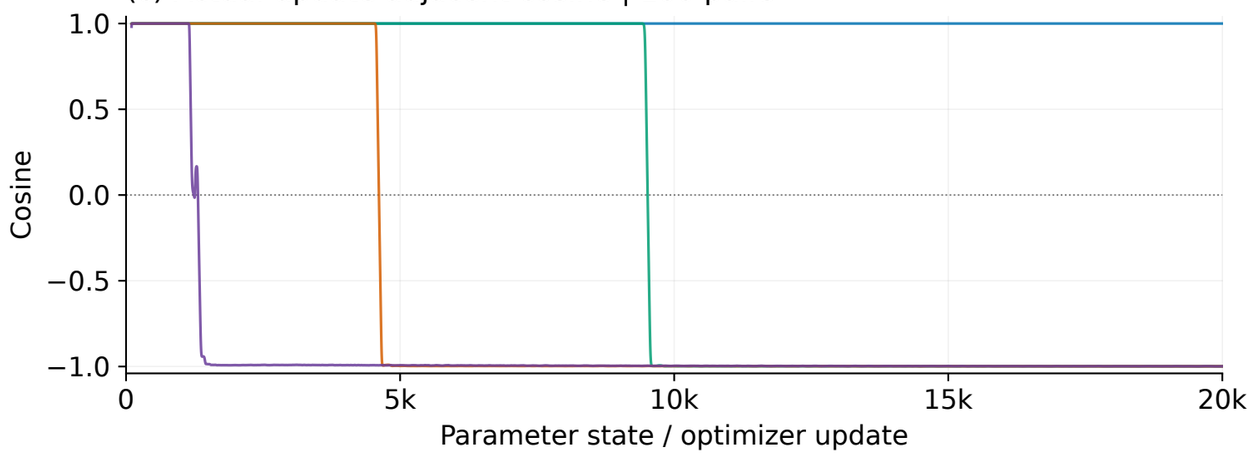
(a) Unsmoothed full-batch training loss



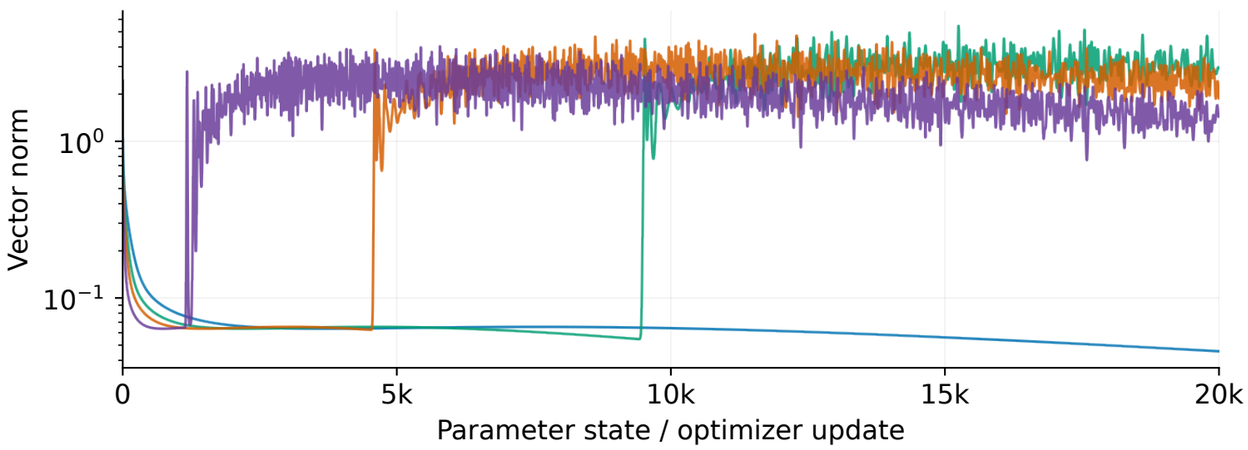
(b) Raw-gradient adjacent cosine | 100 pairs



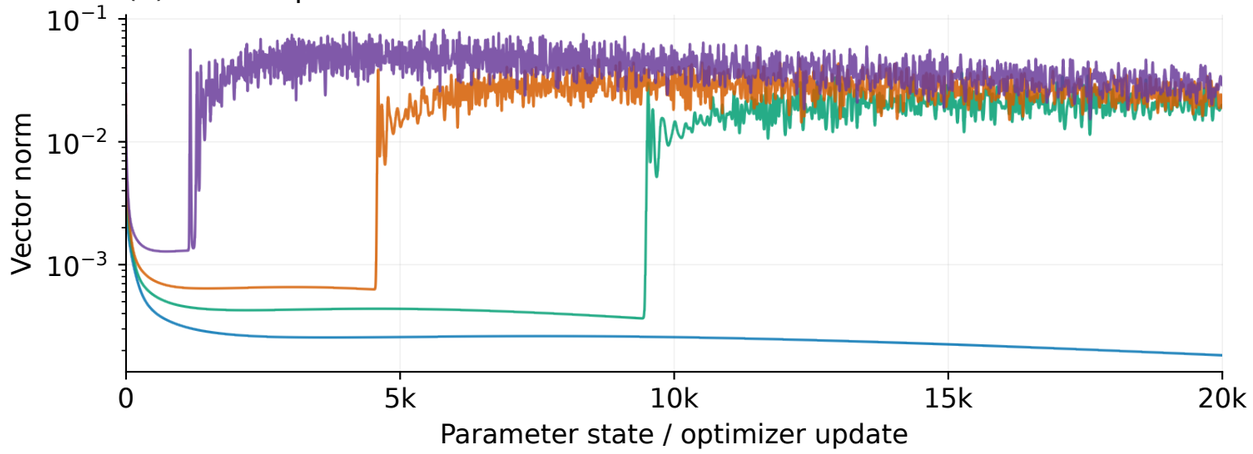
(c) Actual-update adjacent cosine | 100 pairs



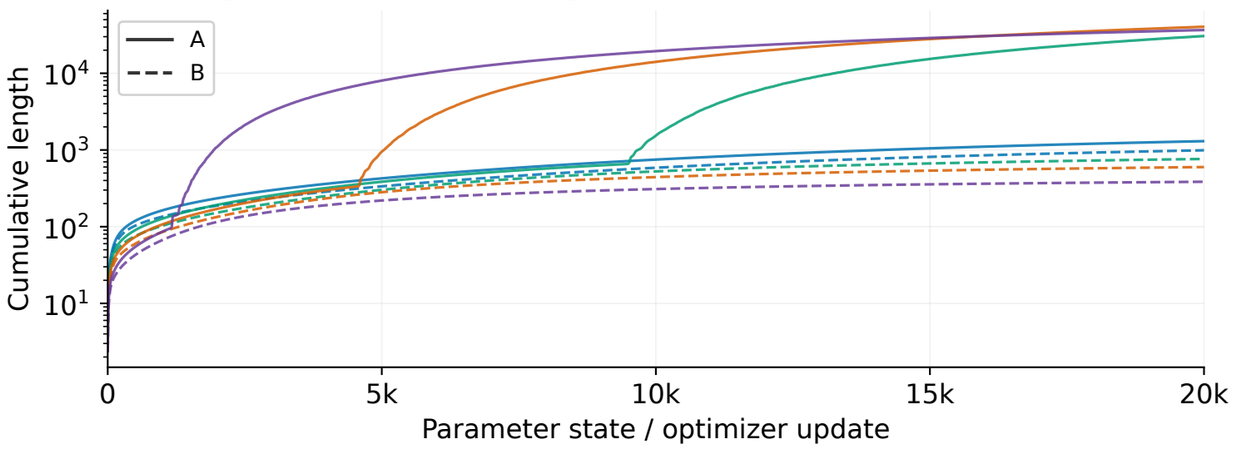
(d) Raw-gradient norm



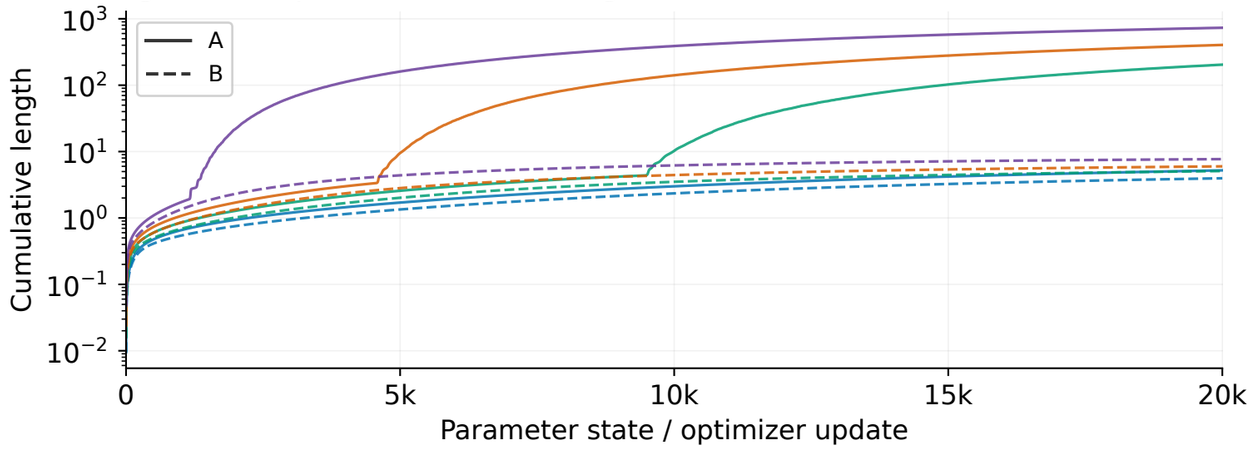
(e) Actual-update norm



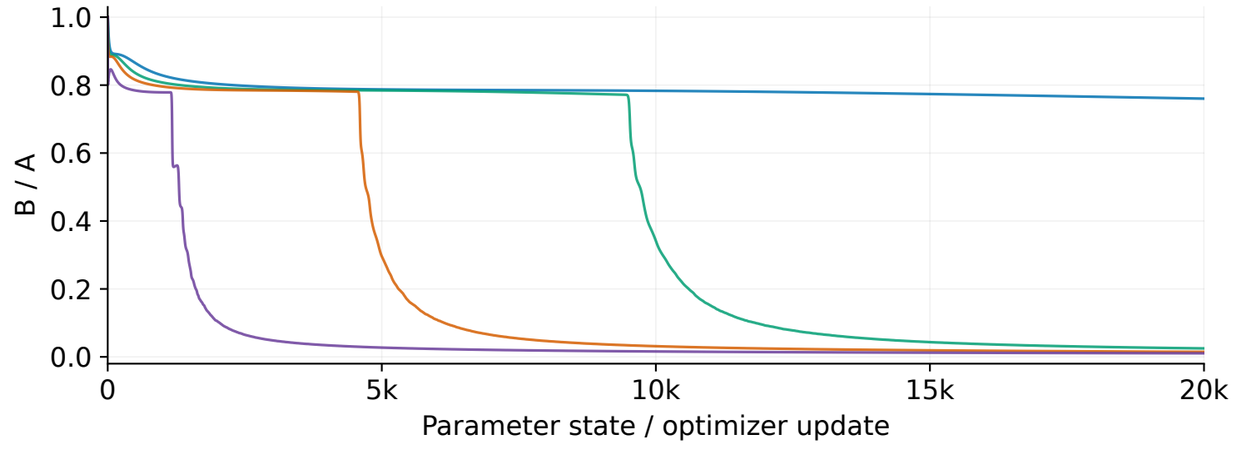
(f) Raw-gradient cumulative lengths



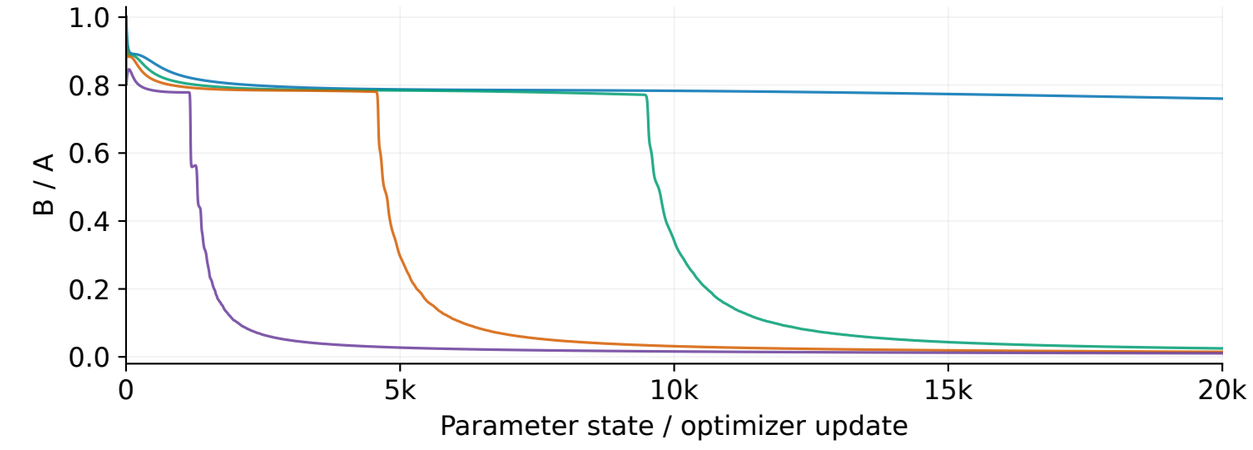
(g) Actual-update cumulative lengths



(h) Raw-gradient cumulative displacement



(i) Actual-update cumulative displacement



Color identifies LR. A=sum of vector norms; B=norm of the vector sum; ratios use the full prefix.
Actual updates include FP32 rounding. All cosine means use 100 complete pairs.